

How I Created a System to Transform Communities into Economies and Ecosystems

Evolving them from being places to connect and engage around shared interests to country-like economies and ecosystems



Undergraduate students studying in an old lecture hall in FUT Minna, Nigeria

Learning is central in how a country or an economy grows and what it can create. The same can be said for communities, and as Barack Obama once said:

"The nation that out-educates us today is going to out-compete us tomorrow."

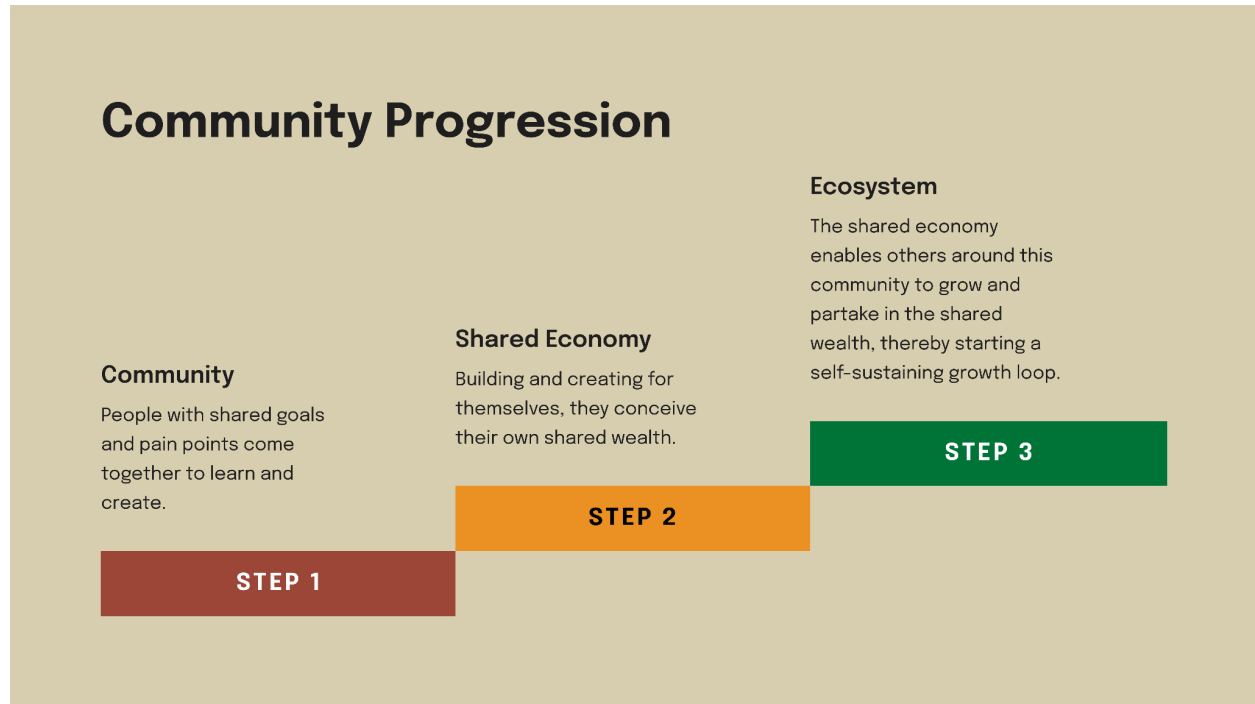
If this is so, how can this be recreated and reapplied to the communities we work in and those around us?

It can be applied to our local communities by enabling the community to:

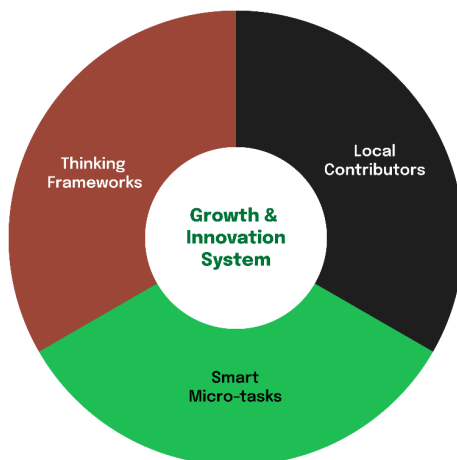
- Learn,
- Engage in and around their learnings, and

- Share the creations of their learnings.

This sharing brings about innovation; it brings about new products, new services and new value systems. These new systems in turn create new markets, and these markets create shared wealth or economy and this enables the community to become an ecosystem.



All of this can be done through certain processes which make up a system I call a Growth and Innovation System.



The GIS expanded:

Thinking Frameworks

- Design Thinking
- Project Thinking
- Market Thinking
- And more...

Local Contributors

- Local community
- Local Knowledge & Culture
- Local Partners

Smart Micro-tasks

- Bite-sized tasks from real world projects/problems
- Summarised guides from open online resources

I stumbled on this while I was working on a completely different project.

In 2016 I was working as a Community Manager for [Devcenter.co](https://devcenter.co) managing their community of developers, designers and other tech enthusiasts who came together to build open source projects but stayed together to connect, help each other and grow.

Within the community, I noticed that developers were learning as they contributed to open source projects and within short time periods, individuals who would ask simple questions were now helping others in the community answer complex questions.

It was amazing to me! I wondered how this individual and learned so happened and whether it could be replicated in a way that enabled mass learning.

Mass learning was important because I had noticed a skill gap in the community. There were those who were skilled, those who were beginners, and no way in between to help the beginners grow. We needed a way to bridge this gap. This led me to do some research.

Researching for answers

As I dug in, I learned there were different forms of learning like Project-based Learning, Task-based Learning, Apprenticeships and a host of other concepts. With all these theories in mind, I decided to create something that allowed me to test them and so I created [Learn by Doing](#).

Learn by Doing

Helping developers learn by contributing to real life projects.

To help reduce the skills gap, and increase the quality of talents available locally, we have created a process around how developers, designers and project managers learn new tools and technologies using open source projects.

This is informed by months of watching various talents work on various projects. We have seen the speed of learning and quality of knowledge gained, when they learn by doing. Research has also shown (see [here here here](#)) that this method of learning helps students gain a deeper understanding of the subject, for a longer time than other traditional means of teaching.

Talents not experienced in a particular skill, will work on projects using specially designed tasks. And with the aid of guides, they will learn new technologies, skills and techniques in software development by working through a list of tasks, which have been set up to help them learn a specific roles/skill areas (like front-end or back-end development, product management or project management) by contributing to a real project.

Look through the projects below, to see its list of tasks, and get a better view, of this journey of learning.

Project List

1. [Coding 101](#)
2. More coming soon.

There were **projects** that had **small size tasks** that carried in them **bite-sized explainers** which were meant to improve one's understanding of the tasks and thus increase their willingness to complete them. There were also mentors who worked one-on-one with the learners on their tasks and they could reach out to these mentors for questions in a private space. And so I had successfully combined all the theories of learning I had found; project-based, task-based, apprenticeships and a host of other learning principles.

Learners were starting the projects and seemed enthusiastic about it, but after 2 weeks or so, they dropped off. Even trying to incentivise their contributions did not work. So I went back to the drawing board and decided to look at how successful learning systems had gone about solving this problem.

Community-based learning

In 2015, Seth Godin founded the [altMBA](#). It's an online leadership and management workshop with a [96% completion rate](#). 96% is key because [studies show the average completion rate of online courses is 4%](#).

The program uses digital tools like Slack, WordPress and Zoom to engage more than 100 students at a time in an intense four-week course. In 2016, students from 27 countries and 85 industries worldwide participated.



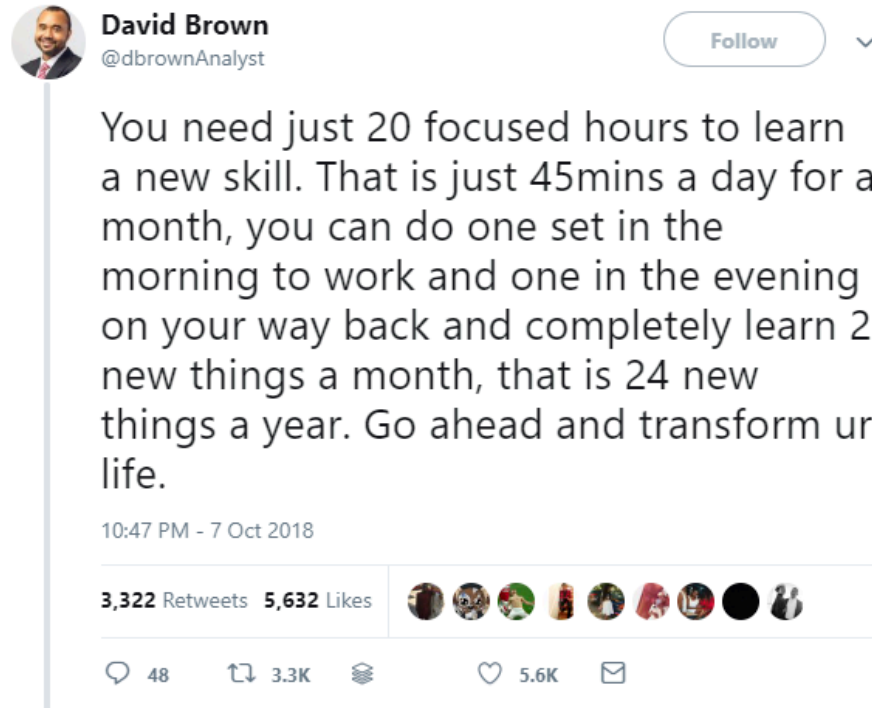
altMBA Students interacting via Zoom - Photo from [Psychology of stuff](#)

Learn anything in 20 hours

In 2013, [Josh Kaufman](#) gave a TEDx talk at the Colorado State University on how to [learn anything in 20 hours](#).

He broke the learning process into 4 steps:

1. Deconstruct the Skill
2. Learn enough to self-correct
3. Remove barriers to practice
4. Practice for at least 20 hours



Screenshot of a [tweet](#) on Josh Kaufman's process

As we can see from [this tweet](#) dated October 2018, Josh Kaufman's process is still in use, it works.

These 2 processes were the ones I saw as the most convincing through my research. I went through quite a lot and I kept seeing similarities through them all. Please see below some of the useful links I found helpful in understanding learning and how others had improved on these processes:

PowToon Animations	https://www.powtoon.com/tutorials/
Open Badges	https://www.youtube.com/watch?v=DiTHluqA0cc
Artificial intelligence & the future of education systems	https://www.youtube.com/watch?v=ZdHhs-l9FVo
Forget university? 4 steps to design your own education	https://www.youtube.com/watch?v=TUnpSYMNEhY
Chicago City of Learning - Creating activities that help individuals earn badges	https://chicagocityoflearning.org/
Growing A Talent Hotbed	https://www.youtube.com/watch?v=Aq0pHpNy6bs
Unboxing education through gaming, playing, and making	https://www.youtube.com/watch?v=CCp_9304j3Q

The first 20 hours -- how to learn anything	https://www.youtube.com/watch?v=5MgBikgcWnY
Can we make learning as much fun as playing a video game?	https://www.youtube.com/watch?v=kr9a_itscg8
21st century learning	https://www.youtube.com/watch?v=UI9TiuVHc0A
Teaching Methods for Inspiring the Students of the Future	https://youtu.be/UCFg9bcW7Bk
Hackschooling makes me happy	https://youtu.be/h11u3vtcpaY

Combining Insights

Next, I tried to combine the insights I had gleaned to create something that reduces the time to learn a skill (TTLS) so that before learners dropped off, they would have gained enough to be valuable.

For this, I used Josh Kaufman's process as a mould to create this new learning process that reduced the TTLS. Josh's process had set the TTLS to be 20 hours, so I believed in building on it or extending it, the TTLS could be reduced.

I used this new process to create a new product or startup I called Frintern; Freelance Interns (now defunct). This explainer was written in the way we used the process to build Frintern.

Deconstructing the skill

First, we started by deconstructing various work skills. For this, we used the first principle. Along with insights from interviews with skilled professionals in various work fields.

The first principle is a way to break down scenarios into basic elements. And then reassemble them from the ground up [via [fs.blog](#)].

This helped us create mental models of these work skills. Mental models are your thought processes about how things work in the real world. They are the most basic form of any skill.

Platforms like [Farnam Street](#) help people learn complex topics fast using mental models. This reduced the Time To Learn A Skill (TTLS) by some amount. Let us call that amount X.

Learning enough to self-correct

To help people learn enough to self-correct, we curated free learning resources and arranged them in a way that let people learn the mental models of various work skills and frameworks. These resources were then broken down into bite sizes and hosted on our microlearning management system. This made learning and assimilation easier.

Learning tools like [Duolingo](#) use microlearning principles to help people learn fast. This also reduced the Time To Learn A Skill (TTLS) by some amount. Let us call that amount Y.

Removing barriers to practice

Fishes are good swimmers, the same goes for people born in riverine areas. If a random riverine dweller decided to become a professional swimmer, it would be easier compared to others. The same is true for mountain dwellers who want to become professional marathon runners.

Using this pattern, we built the personalisation system for learning and career paths.

Personalising the learning experience removes a lot of learning barriers, and it reduced the Time To Learn A Skill (TTLS) by some amount. Let us call that amount C.

Practising for at least 20 hours

To allow individuals to practice fast, we designed [Learning Labs and Sprints](#). These allow practising these career skills using micro-content, micro-tasks and micro-interactions.

Learning in sprints also prepares the learners for real-world work experience. Especially as they work on real-world projects in these sprints. Here we bring in the community-based learning methodology.

Learning labs and sprints also improve execution and learning speed by some amount. Let us call that amount D.

The New Time To Learn A Skill (TTLS)

Based on the explanations above, the Time To Learn A Skill (TTLS) becomes:

$$\text{TTLS} = \text{TTLS} - (X + Y + C + D)$$

Such that whatever the new TTLS is, it is much shorter than 20 hours.

In our [early pilot workshops that ran from March to May 2018](#), we found that people could learn a skill and use it to create something in 2 - 3 hours. That put our TTLS at 2 - 3 hours.

What Next?

After spending months synthesising insights from all of these, testing the process with teams and individuals, I realised I had created something much more than a way to help people learn and reduce unemployment. I realised we had created a way for people to build businesses via projects while learning in a community and it all happened in a way where it was student or community-led and so agency rested with the learners and their success was a direct result of their contributions.

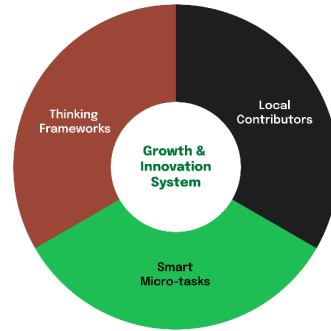
After explaining it to [Sean Burrowes](#), he called it The Innovation Engine. I call it a Growth and Innovation System.

A Growth and Innovation System

What is it?

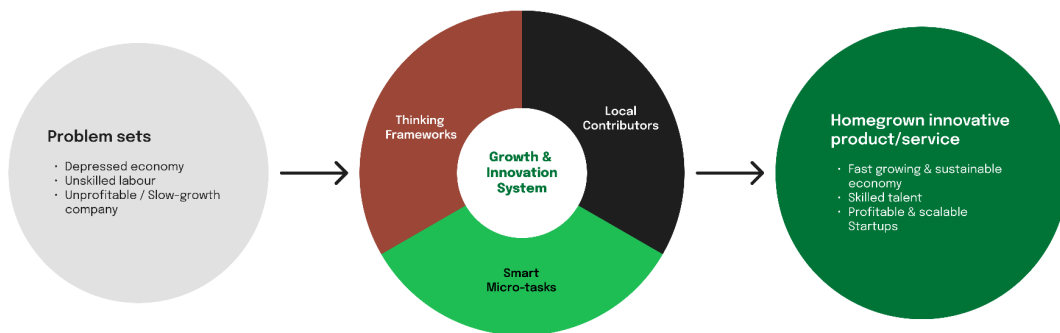
The Growth and Innovation system

The Growth and Innovation System (GIS) is a process that facilitates skill and resource crowdsourcing to enable communities to solve their problems themselves while creating community wealth or prosperity.



Explaining/Breaking down the System

Members of a local community with a shared pain point go through various thinking workshops. These workshops help them understand different new thinking processes and frameworks. With these frameworks, they are better able to convert their problems into innovative solutions that allow create shared economies and ecosystems.

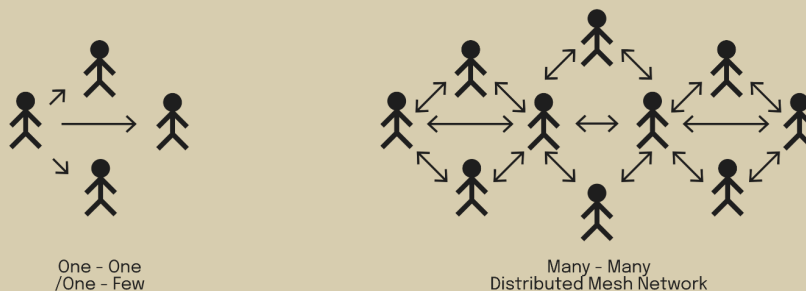


Problem sets are examined through the GIS and they become Homegrown innovative products and services

Another key part of this process is that it helps makes learning, researching, ideation which has traditionally happened on one-to-one or one-to-many connections, happen on a many-to-many connection, which increases how these happen exponentially because of the multiple contributors.

Innovation, Research, Ideation, Learning, always happened on a one-one or one-few relationship/connection.

Imagine, if now, it could happen on a many-many relationship/connection. Like a mesh network. A distributed mesh network.



Taking it a step further: Automating the system

I tried to automate this process or system at frintern.com (now techstarta.com), but back then I couldn't properly explain it. Before I left Frintern to my cofounder Jide Owosakin, we were able to:

- digitise the learning process using scripted programs allowing it to happen virtually via any channel, text, audio or video.
- design, implement and test a basic personalisation system that leverages an individual's affinities to create connections to skills they'll naturally be good at.
- design and implement a microlearning management system that enables the self-learning of work-based mental models.

And all of these systems were built distributedly and they worked interconnectedly. If you go on techstarta.com, you should still see the basic form.

The systems were proof they could be digitised, but we were missing a key part, a place for the many to many interactions to happen virtually.

How could we do what we had [successfully done within a physical community](#) in a virtual community? How could we enable the many-to-many interactions and engagements to happen in a virtual space such that the innovations and markets could be created virtually?

This question led me to Discourse.org. The way Discourse is built allows extensibility through plugins and theme components. It is a community platform that is also able to do everything a content management system can do.

This meant I would be able to recreate the tools we had built at Frintern on top of Discourse using its plugin and theme component system while leveraging the community features of Discourse. I am currently testing this out at community.growthclinic.xyz.

It is still in its very very early stages, but it is the same idea (even if it doesn't seem this way) implemented in the most basic form that Discourse allows; coordinated learning and sharing between multiple individuals.

Doing it this way will allow me to learn from the users what they expect the system/tool to be, so I can build exactly what they need.